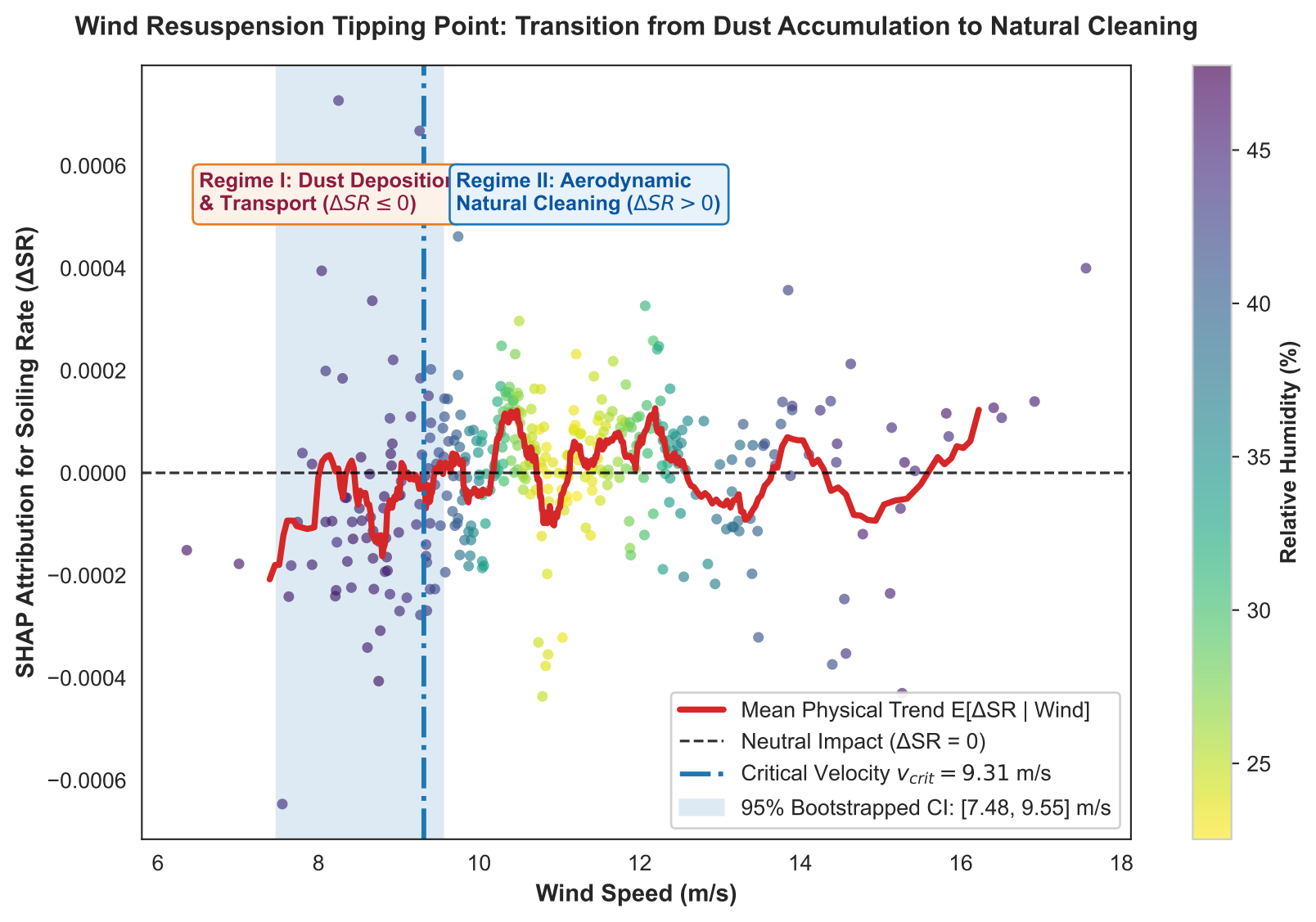

Wind Resuspension Tipping Point: Transition from Dust Accumulation to Natural Cleaning
Regime I: Dust Deposition & Transport (ΔSR ≤ 0)
Regime II: Aerodynamic Natural Cleaning (ΔSR > 0)
Mean Physical Trend E[ΔSR | Wind]
Neutral Impact (ΔSR = 0)
Critical Velocity v_{crit} = 9.31 m/s
95% Bootstrapped CI: [7.48, 9.55] m/s
SHAP Attribution for Soiling Rate (ΔSR)
Wind Speed (m/s)
Relative Humidity (%)
0.0006
0.0004
0.0002
0.0000
−0.0002
−0.0004
−0.0006
6
8
10
12
14
16
18
45
40
35
30
25